

one model, one forward pass, two predictions

check-in window
(*semantic stream*)

semantic encoder
(*private*)

region window
(*spatial stream*)

spatial encoder
(*private*)

shared trunk (bidirectional cross-attention $\times 2$)

← streams exchange information →

category output

→ *next-category*

region output
shared + private spatial path

→ *next-region*

raw region
(*private
spatial path*)